

Year 7 · Chapter 5A · Algebra: answers (teacher copy)

5.01 The commutative and associative laws (booklet p. 3)

Examples (model these live)

E1 $25 \times 17 \times 4$
 $= 25 \times 4 \times 17 \cdot = 100 \times 17 = 1\,700$

E2 $38 + 47 + 62$
 $= 38 + 62 + 47 \cdot = 100 + 47 = 147$

E3 Is $20 - 8$ the same as $8 - 20$?
 $20 - 8 = 12, 8 - 20 = -12 \cdot$ no:
subtraction is not commutative

Questions (43 on today's skill, 17 from earlier lessons)

- | | | | |
|---|--|--|---|
| 1 1 400 | 20 6×9 ; yes, both are 54 | 34 $= 50 \times 2 \times 18 \cdot = 100 \times 18 \cdot$
$= 1\,800$ | 47 $= (25 \times 4) \times (3 \times 5) \cdot = 100$
$\times 15 \cdot = 1\,500$ |
| 2 122 | 21 0.26 4.09 | 35 $\times 100: 20\% \cdot \frac{20}{100} = \frac{1}{5}$ 4.10 | 48 $\frac{3}{2}$ 4.01 |
| 3 6×11 ; yes, both are 66 | 22 2 700 | 36 $= (5 \times 2) \times (6 \times 6) \cdot = 10 \times$
$36 \cdot = 360$ | 49 commutative law |
| 4 900 | 23 $= 125 \times 8 \times 6 \cdot = 1000 \times 6 \cdot$
$= 6\,000$ | 37 $= 31 + 69 + 45 + 21 \cdot = 100$
$+ 66 \cdot = 166$ | 50 HCF $= 18 \cdot = 3/4$ 3.11 |
| 5 112 | 24 $= 26 + 74 + 24 + 25 \cdot = 100$
$+ 49 \cdot = 149$ | 38 10 4.07 | 51 associative law |
| 6 12×8 ; yes, both are 96 | 25 $\frac{1}{3}$ 4.02 | 39 Yes: both are 15
(associative law). | 52 $= 25 \times 2 \times 44 \cdot = 50 \times 44 =$
2 200 |
| 7 300 | 26 No: 3 and 12. Division is
not associative. | 40 $= \frac{7}{2} \times \frac{2}{5} \cdot = 1 \frac{2}{5}$ 4.06 | 53 $\frac{2}{3}$ 4.12 |
| 8 78 | 27 $= 5 \times 2 \times 18 \cdot = 10 \times 18 \cdot =$
180 | 41 $= (20 \times 5) \times (6 \times 8) \cdot = 100$
$\times 48 \cdot = 4\,800$ | 54 $= (132 + 68) + (79 + 21) \cdot =$
$200 + 100 = 300$ |
| 9 65.1 4.11 | 28 $= 7 + 9 \cdot = 16$ 3.08 | 42 associative law | 55 11, 13, 17, 19, 23, 29 3.09 |
| 10 3×10 ; yes, both are 30 | 29 $= 23 + 77 + 31 + 83 \cdot = 100$
$+ 114 \cdot = 214$ | 43 2×3^3 3.10 | 56 $= (25 \times 4) \times (19 \times 2) \cdot = 100$
$\times 38 \cdot = 3\,800$ |
| 11 2 300 | 30 Yes: both are 42
(commutative law). | 44 $= 15 \times 2 \times 14 \cdot = 30 \times 14 =$
420 | 57 No: 3 and $\frac{1}{3}$. Division is
not commutative. |
| 12 111 | 31 $= 37 + 63 + 68 + 75 \cdot = 100$
$+ 143 \cdot = 243$ | 45 last two digits: 28 · yes: 28
is divisible by 4 3.04 | 58 28 4.05 |
| 13 5×8 ; yes, both are 40 | 32 No: 3 and 9. | 46 $= (101 + 99) + (55 + 45) \cdot =$
$200 + 100 = 300$ | 59 commutative law |
| 14 92 | 33 50 3.01 | | 60 $C + X + VI \cdot 100 + 10 + 6 =$
116 3.06 |
| 15 10×3 ; yes, both are 30 | | | |
| 16 300 | | | |
| 17 7 3.03 | | | |
| 18 $= 125 \times 8 \times 12 \cdot = 1000 \times 12$
$\cdot = 12\,000$ | | | |
| 19 145 | | | |

5.02 The distributive law (booklet p. 5)

Examples (model these live)

E1 7×43
 $= 7 \times 40 + 7 \times 3 \cdot = 280 + 21 = 301$

E2 9×26
 $= 10 \times 26 - 1 \times 26 \cdot = 260 - 26 = 234$

E3 Expand $5(x + 3)$.
 $= 5 \times x + 5 \times 3 \cdot = 5x + 15$

Questions (40 on today's skill, 17 from earlier lessons)

- | | | | |
|---|--|--|---|
| 1 $= 4 \times 30 + 4 \times 4 \cdot = 120 +$
16 = 136 | 15 $3 \times 18 = 54 \cdot 30 + 24 = 54$ | 31 No: 3 and 12. Division is
not associative. 5.01 | 45 8×11 ; yes, both are 88
5.01 |
| 2 $9 \times 59 = 531 \cdot 450 + 81 =$
531 | 16 $9x + 90$ | 32 $= 3 \times 3a + 3 \times 4 \cdot = 9a + 12$ | 46 $= 6 \times 50 - 6 \times 1 \cdot = 300 - 6 \cdot$
$= 294$ |
| 3 $4m + 48$ | 17 5×9 ; yes, both are 45 5.01 | 33 $= 2 \times y - 2 \times 9 \cdot = 2y - 18$ | 47 $= 5 \times 2 \times 6 \cdot = 10 \times 6 \cdot = 60$
5.01 |
| 4 $= 8 \times 40 + 8 \times 3 \cdot = 320 +$
24 = 344 | 18 $= 7 \times 40 + 7 \times 7 \cdot = 280 +$
49 = 329 | 34 HCF of 48 and 52 is 4 · =
$\frac{12}{13}$ 4.01 | 48 $= 6 \times 200 - 6 \times 1 \cdot = 1\,200$
$- 6 \cdot = 1\,194$ |
| 5 $8 \times 51 = 408 \cdot 400 + 8 =$
408 | 19 $= 20 \times 34 - 1 \times 34 \cdot = 680 -$
34 · = 646 | 35 $\square = 1$ | 49 No. Multiply both terms:
$2b + 16$. |
| 6 $2x + 8$ | 20 $= 3 \times c - 3 \times 6 \cdot = 3c - 18$ | 36 2 300 5.01 | 50 CCC + X + VI · 300 + 10 + 6
$= 316$ 3.06 |
| 7 $= 3 \times 30 + 3 \times 4 \cdot = 90 + 12$
$= 102$ | 21 $= 20 \times 5 \times 4 \cdot = 100 \times 4 \cdot =$
400 5.01 | 37 $= 10 \times 59 - 1 \times 59 \cdot = 590 -$
59 · = 531 | 51 $= 4 \times (28 + 26) \cdot = 4 \times 54 =$
216 |
| 8 $3 \times 45 = 135 \cdot 120 + 15 =$
135 | 22 $\square = 7$ | 38 $= 5 \times 5t - 5 \times 8 \cdot = 25t - 40$ | 52 114 5.01 |
| 9 110 5.01 | 23 $= 30 \times 30 - 1 \times 30 \cdot = 900 -$
30 · = 870 | 39 $= 3 \times 1000 - 3 \times 1 \cdot = 3\,000$
$- 3 \cdot = 2\,997$ | 53 $= 5 \times 2p - 5 \times 4 \cdot = 10p -$
20 |
| 10 $6 \times 33 = 198 \cdot 180 + 18 =$
198 | 24 $6 \frac{1}{5}$ 4.04 | 40 $= 40 + 60 + 29 + 17 \cdot = 100$
$+ 46 \cdot = 146$ 5.01 | 54 $\square = 9$ |
| 11 $4n + 12$ | 25 $= 5 \times t - 5 \times 10 \cdot = 5t - 50$ | 41 No. Multiply both terms:
$4n + 12$. | 55 Yes: both are 42
(commutative law). 5.01 |
| 12 $= 5 \times 20 + 5 \times 9 \cdot = 100 +$
45 = 145 | 26 $\square = 5$ | 42 4 4.05 | 56 $= 8 \times 50 - 8 \times 1 \cdot = 400 - 8$
$\cdot = 392$ |
| 13 $\frac{1}{7}$ 4.02 | 27 $= 2 \times k - 2 \times 7 \cdot = 2k - 14$ | 43 $= 9 \times (26 + 21) \cdot = 9 \times 47 =$
423 | 57 2×3^3 3.10 |
| 14 $= 20 \times 46 - 1 \times 46 \cdot = 920 -$
46 · = 874 | 28 $\square = 6$ | 44 $= 5 \times 5x - 5 \times 9 \cdot = 25x - 45$ | |
| | 29 131 5.01 | | |
| | 30 $= 10 \times 18 - 1 \times 18 \cdot = 180 -$
18 · = 162 | | |

5.03 Variables and algebraic notation (booklet p. 7)

Examples (model these live)

E1 Simplify $5 \times y$.
number first $\cdot = 5y$

E2 Simplify $b \times 3 \times a$.
 $= 3 \times a \times b \cdot = 3ab$

E3 Matchstick squares in a row: 1 square uses 4, 2 squares use 7, 3 squares use 10. Write a rule.
each new square adds 3 $\cdot m = 3s + 1$

Questions (42 on today's skill, 17 from earlier lessons)

- | | | | |
|--|---|--|--|
| 1 $10a$ | 19 $9 \times 12 = 108 \cdot 90 + 18 = 108$
5.02 | 35 $= 5 \times 2 \times b \times b \times n \cdot =$
$10b^2n$ | 47 $8 \times 17 = 136 \cdot 80 + 56 = 136$
5.02 |
| 2 t^2 | 20 $3b$ | 36 $D + XL + V \cdot 500 + 40 + 5 =$
545 3.06 | 48 $t = 5 \times 14 + 1 \cdot = 71$ |
| 3 $4 \times p$ | 21 $5cx$ | 37 $3 \times m \times t$ | 49 Yes: both are 15
(associative law). 5.01 |
| 4 $4a$ | 22 $4 \times a \times k$ | 38 $= 3 \times 40 + 3 \times 4 \cdot = 120 +$
$12 = 132$ 5.02 | 50 $t = 3 \times 9 + 4 \cdot = 31$ |
| 5 c^2 | 23 No: 3 and 9. 5.01 | 39 $\frac{m}{6}$ | 51 y goes up by 4: $4x \cdot y = 4x$
$+ 3$ |
| 6 $12 \times m$ | 24 $\frac{t}{6}$ | 40 $= 3 \times 5 \times n \times n \times p \cdot =$
$15n^2p$ | 52 short division, left to right
$\cdot = 175$ 3.03 |
| 7 $3k$ | 25 goes up by 2 each time $\cdot$
term $= 2n + 2$ | 41 $t = 2 \times 16 + 4 \cdot = 36$ | 53 No. Both mean $k \times p$; we
write kp . |
| 8 a^2 | 26 22; no 3.04 | 42 $= 16 + 84 + 65 + 53 \cdot = 100$
$+ 118 \cdot = 218$ 5.01 | 54 $2a + 20$ 5.02 |
| 9 107 5.01 | 27 $3kp$ | 43 y goes up by 4: $4x \cdot y = 4x$
$+ 4$ | 55 $= 6 \times 5 \times b \times b \times c \cdot = 30b^2c$ |
| 10 $8 \times n$ | 28 $9 \times k \times t$ | 44 200 3.01 | 56 goes up by 4 each time $\cdot$
term $= 4n + 6$ |
| 11 $7y$ | 29 $5 \times p \times x$ | 45 No. Both mean $t \times y$; we
write ty . | 57 $= 5 \times 2 \times 14 \cdot = 10 \times 14 \cdot =$
140 5.01 |
| 12 p^2 | 30 $\frac{m}{5}$ | 46 $= 3 \times 2 \times a \times a \times m \cdot =$
$6a^2m$ | 58 $t = 5 \times 20 + 3 \cdot = 103$ |
| 13 $11 \times k$ | 31 $4p + 44$ 5.02 | | 59 18 4.05 |
| 14 $3t$ | 32 goes up by 3 each time $\cdot$
term $= 3n + 5$ | | |
| 15 $1 \times 10, 2 \times 5 \cdot 1, 2, 5, 10$
3.09 | 33 $= 50 \times 2 \times 13 \cdot = 100 \times 13 \cdot$
$= 1300$ 5.01 | | |
| 16 $7bx$ | 34 $4ny$ | | |

5.04 Expressions from words (booklet p. 9)

Examples (model these live)

E1 5 more than n
add 5 $\cdot = n + 5$

E2 3 less than twice x
twice x : $2x \cdot = 2x - 3$

E3 The cost of p pens at \$4 each
 $4 \times p \cdot = 4p$ dollars

Questions (40 on today's skill, 17 from earlier lessons)

- | | | | |
|---|--|---|---|
| 1 $a + 4$ | 18 $9a$ | 34 $CC + L \cdot 200 + 50 = 250$ 3.06 | 47 $= 125 \times 8 \times 4 \cdot = 1000 \times 4 \cdot$
$= 4000$ 5.01 |
| 2 $k - 5$ | 19 first: $n^2 \cdot = n^2 + 2$ | 35 first: $bp \cdot = bp - 8$ | 48 spiders: $8t$, birds: $2 \times 8 =$
$16 \cdot = 8t + 16$ |
| 3 $6a$ | 20 first: $5k \cdot = 5k + 2$ | 36 11×5 ; yes, both are 55
5.01 | 49 $\frac{110}{p}$ dollars |
| 4 $\frac{b}{8}$ | 21 $= 11 + 89 + 31 + 84 \cdot = 100$
$+ 115 \cdot = 215$ 5.01 | 37 first: $a + 4 \cdot = 9(a + 4)$ | 50 list multiples of the
largest; test the others $\cdot$ LCM $=$
60 3.12 |
| 5 $a + 11$ | 22 first: $6n \cdot = 6n - 4$ | 38 $m =$ number of pens $\cdot =$
$13m + 16$ dollars | 51 double the sum of b and
5 |
| 6 $2k$ | 23 first: $a + m \cdot = \frac{a + m}{7}$ | 39 spiders: $8x$, birds: $2 \times 11 =$
$22 \cdot = 8x + 22$ | 52 a^2 5.03 |
| 7 b^2 | 24 $1\frac{1}{5}$ 4.06 | 40 goes up by 3 each time $\cdot$
term $= 3n + 2$ 5.03 | 53 $m =$ number of pens $\cdot =$
$2m + 5$ dollars |
| 8 $c - 10$ | 25 first: $km \cdot = km - 9$ | 41 $\frac{110}{y}$ dollars | 54 first: $2b \cdot = 2b - 11$ |
| 9 c^2 5.03 | 26 first: $c + 9 \cdot = 9(c + 9)$ | 42 25 4.05 | 55 $= 9 \times m - 9 \times 8 \cdot = 9m - 72$
5.02 |
| 10 $m + 10$ | 27 first: $2a \cdot = 2a - 4$ | 43 double the sum of m and
5 | 56 spiders: $8x$, birds: $2 \times 3 = 6$
$\cdot = 8x + 6$ |
| 11 bm | 28 first: $6b \cdot = 6b + 4$ | 44 $a =$ number of pens $\cdot = 3a$
$+ 6$ dollars | 57 61 3.03 |
| 12 $t + 4$ | 29 $10 \times a$ 5.03 | 45 $= 8 \times 30 + 8 \times 4 \cdot = 240 +$
$32 = 272$ 5.02 | |
| 13 yes 3.11 | 30 first: $8a \cdot = 8a - 9$ | 46 spiders: $8b$, birds: $2 \times 4 =$
$8 \cdot = 8b + 8$ | |
| 14 first: $2b \cdot = 2b - 3$ | 31 $\square = 8$ 5.02 | | |
| 15 $\frac{n}{8}$ | 32 $n =$ number of pens $\cdot =$
$14n + 17$ dollars | | |
| 16 $p - 7$ | 33 first: $m^2 \cdot = m^2 + 6$ | | |
| 17 $6 \times 34 = 204 \cdot 180 + 24 =$
204 5.02 | | | |

5.05 Substitution (booklet p. 11)

Examples (model these live)

E1 Find $4m - 3$ when $m = 6$.
 $= 4 \times 6 - 3 \cdot = 21$

E2 Find $ab + b$ when $a = 3$ and $b = 5$.
 $= 3 \times 5 + 5 \cdot = 20$

E3 $A = lw$. Find A when $l = 8$ and $w = 5$.
 $A = 8 \times 5 \cdot = A = 40$

Questions (40 on today's skill, 17 from earlier lessons)

1	14	17	n^2 5.03	32	$C = 50 \times 5 + 25 \cdot C = \275	45	$= 25 \times 4 \times 19 \cdot = 100 \times 19 \cdot$
2	64	18	73	33	$P = 2 \times 11 + 2 \times 5 \cdot P = 32$		$= 1\ 900$ 5.01
3	7	19	$= 2 \times 2 + 9 \cdot = 13$	34	$8c$ 5.04	46	$v = 9 + 2 \times 4 \cdot v = 17$
4	12	20	$P = 2 \times 10 + 2 \times 5 \cdot P = 30$	35	$= 4 \times (-6) + 4 \cdot = -20$	47	first: $2c \cdot = 2c - 4$ 5.04
5	16	21	$= 36 + 64 + 28 + 13 \cdot = 100$	36	$\frac{f}{4}$ 5.03	48	$v = 5 + 2 \times 10 \cdot v = 25$
6	6		$+ 41 \cdot = 141$ 5.01	37	$= 5 \times 6 - 8 \cdot = 22$	49	$= (-4)^2 - 3 \times 6 \cdot = 16 - 18 \cdot$
7	17	22	$= 5 \times (-5) + 4 \cdot = -21$	38	$C = 56 \times 2 + 20 \cdot C = \132		$= -2$
8	81	23	$= 3 \times 2 - 6 \cdot = 0$	39	$v = 10 + 5 \times 6 \cdot v = 40$	50	$1 \times 18, 2 \times 9, 3 \times 6 \cdot 1, 2, 3,$
9	$m - 4$ 5.04	24	$= 4 \times c - 4 \times 8 \cdot = 4c - 32$	40	short division, left to right		$6, 9, 18$ 3.09
10	16		5.02		$\cdot = 193$ 3.03	51	No. Square first: $3 \times 5^2 = 3$
11	2	25	$P = 2 \times 10 + 2 \times 4 \cdot P = 28$	41	$= (-1)^2 - 3 \times 4 \cdot = 1 - 12 \cdot =$		$\times 25 = 75.$
12	10	26	$= 2 \times (-6) + 3 \cdot = -9$		-11	52	105 5.01
13	8 3.05	27	$P = 2 \times 5 + 2 \times 12 \cdot P = 34$	42	$9t + 72$ 5.02	53	$C = 38 \times 8 + 33 \cdot C = \337
14	$= 4 \times 8 \cdot = 32$	28	$= 5 \times (-1) + 3 \cdot = -2$	43	No. Square first: $3 \times 6^2 = 3$	54	$= 5 \times (-1) + 6 \cdot = 1$
15	9	29	4 3.07		$\times 36 = 108.$	55	8t 5.03
16	3	30	$= 5 + 2 \times 3 \cdot = 11$	44	$C = 57 \times 6 + 17 \cdot C = \359	56	$v = 11 + 2 \times 4 \cdot v = 19$
		31	6×9 ; yes, both are 54 5.01			57	$= 30 \times 41 - 1 \times 41 \cdot = 1\ 230$
							$- 41 \cdot = 1\ 189$ 5.02

5.07 Adding and subtracting terms (booklet p. 13)

Examples (model these live)

E1 $5x + 3x$
like terms $\cdot = 8x$

E2 $7a + 2b - 3a$
 $= 7a - 3a + 2b \cdot = 4a + 2b$

E3 Are $3xy$ and $5yx$ like terms?
 yx is the same as $xy \cdot$ yes

Questions (40 on today's skill, 17 from earlier lessons)

1	12t	20	$= 4p + 9y + 4p + 9y \cdot = 8p +$	33	$= 4t + 3y + 4t + 3y \cdot = 8t +$	46	$= 9c - 15c - 3y + 4y \cdot = -6c$
2	13a		18y		6y		+ y
3	no	21	$= 19 + 81 + 13 + 27 \cdot = 100$	34	4 5.05	47	$= 5 \times 9 \cdot = 45$ 5.05
4	16n		$+ 40 \cdot = 140$ 5.01	35	$= 8p + 7p + 6 \cdot = 15p + 6$	48	$= 7a - 10a - 7x + 4x \cdot =$
5	2x	22	$= 4c + 7c + 4 \cdot = 11c + 4$	36	$\frac{y}{2}$ 5.03		$-3a - 3x$
6	no	23	$= 8k + 7k + 7x - x \cdot = 15k +$	37	$= 2a + 6a + 4p + 5p \cdot = 8a +$	49	No. 4b and 4c are not
7	11p		6x		9p		like terms, so it stays $4b + 4c.$
8	7p	24	first: $8t \cdot = 8t + 5$ 5.04	38	$= 7cx - 6cx + 3c^2 + 6c^2 + 9$	50	$\frac{1}{7}$ 4.02
9	16 5.05		$+ 10x$		$\cdot = cx + 9c^2 + 9$	51	$= 11c \cdot = 11 \times 4 = 44$
10	6x	26	$= 3a + 6a + 8 \cdot = 9a + 8$	39	$= 2b - 13b - 6n + 3n \cdot =$	52	$7x + 49$ 5.02
11	no		$+ 12p$		$-11b - 3n$	53	$= 9cp - cp + 6c^2 + 7c^2 + 3 \cdot$
12	18n	27	$= 7b + 6p + 7b + 6p \cdot = 14b$	40	$= \frac{7}{2} \times \frac{1}{2} \cdot = 1\frac{3}{4}$ 4.06		$= 8cp + 13c^2 + 3$
13	7 3.05	28	$= 3b + 3b + 3 \cdot = 6b + 3$	41	No. 5c and 4m are not	54	$= 9y + y + 9 \cdot = 10y + 9$
14	$= 6c + 7c + 7m + m \cdot = 13c$	29	2 3.08		like terms, so it stays $5c + 4m.$	55	6t 5.03
15	5t	30	$= 7b + b + 3y + y \cdot = 8b +$	42	5b 5.04	56	$= 7m - 10m - 7y + 8y \cdot =$
16	no		4y	43	$= 14b \cdot = 14 \times 4 = 56$		$-3m + y$
17	a^2 5.03	31	$9 \times 12 = 108 \cdot 90 + 18 = 108$	44	$= 6bk - 4bk + 8b^2 + b^2 + 6 \cdot$	57	first: $2p \cdot = 2p - 12$ 5.04
18	15k		5.02		$= 2bk + 9b^2 + 6$		
19	$= 4m + 5m + 8y - 6y \cdot = 9m$	32	$= 6am - am + 6a^2 + 3a^2 +$	45	No. 3 and $\frac{1}{3}$. Division is		
	+ 2y		$3 \cdot = 5am + 9a^2 + 3$		not commutative. 5.01		

5.08 Multiplying terms (booklet p. 15)

Examples (model these live)

E1 $4 \times 3y$
 $4 \times 3 = 12 \cdot = 12y$

E2 $5a \times 2b$
 $5 \times 2 = 10, a \times b = ab \cdot = 10ab$

E3 $3m \times (-2m)$
 $3 \times (-2) = -6, m \times m = m^2 \cdot = -6m^2$

Questions (49 on today's skill, 17 from earlier lessons)

1	16n	10	$2y^2$	19	32m	28	$= 6 \times 2 \times ky \times ky \cdot = 12k^2y^2$
2	28mx	11	48c	20	49 3.07	29	$= 2t \times 9y \cdot = 18ty\text{cm}^2$
3	$7n^2$	12	20nt	21	$= 6 \times 3 \times p \times t \cdot = 18pt$	30	$= 14 + 86 + 58 + 37 \cdot = 100$
4	18a	13	$6x^2$	22	30my		$+ 95 \cdot = 195$ 5.01
5	20mn	14	10k	23	$3b^2$	31	$2 \times 2 = 4 \cdot = 4x^2$
6	$5p^2$	15	12cm	24	t^2 5.03	32	$= 4 \times 2 \times ac \times a \cdot = 8a^2c$
7	42n	16	$4y^2$	25	10c	33	$P = 2 \times 11 + 2 \times 5 \cdot P = 32$
8	12nt	17	12ty				5.05
9	4m 5.07	18	$9t^2$	26	$= 3 \times 4 \times c \times n \cdot = 12cn$	34	$= 3n \times 7t \cdot = 21nt\text{cm}^2$
				27	15bn		

$$\begin{aligned}
35 \quad & 2 \times 2 = 4 \cdot = 4y^2 \\
36 \quad & = 7b \times 2p \cdot = 14bp \text{ cm}^2 \\
37 \quad & 4 \times 4 = 16 \cdot = 16t^2 \\
38 \quad & 26 \text{ 4.11} \\
39 \quad & = 5 \times 3 \times c \times ck \cdot = 15c^2k \\
40 \quad & 7 \times 43 = 301 \cdot 280 + 21 = \\
301 \text{ 5.02} \\
41 \quad & \text{sign: } -; 3 \times 5 \times 3 = 45 \cdot = \\
-45t^2y \\
42 \quad & = 3m \times 4x \cdot = 12mx \text{ cm}^2 \\
43 \quad & k - 5 \text{ 5.04}
\end{aligned}$$

$$\begin{aligned}
44 \quad & 4 \times 4 = 16 \cdot = 16a^2 \\
45 \quad & = 5y + y + 7 \cdot = 6y + 7 \text{ 5.07} \\
46 \quad & = 5 \times 6 \times np \times n \cdot = 30n^2p \\
47 \quad & \text{sign: } +; 3 \times 6 \times 3 = 54 \cdot = \\
54b^2p \\
48 \quad & 3 \times 3 \times 3 = 27 \cdot = 27a^3 \\
49 \quad & 5 + 2 = 7 \cdot \frac{3}{12} + \frac{8}{12} = \frac{11}{12} \cdot = \\
7\frac{11}{12} \text{ 4.04} \\
50 \quad & \text{No. } a \times a = a^2, \text{ so it is } \\
12a^2.
\end{aligned}$$

$$\begin{aligned}
51 \quad & 5 \times x \text{ 5.03} \\
52 \quad & \text{sign: } -; 6 \times 2 \times 2 = 24 \cdot = \\
-24b^2m \\
53 \quad & 2 \times 2 \times 2 = 8 \cdot = 8b^5 \\
54 \quad & \text{No: 3 and 9. 5.01} \\
55 \quad & \text{No. } k \times k = k^2, \text{ so it is } 30k^2. \\
56 \quad & = 3 \times (-2) + 3 \cdot = -3 \text{ 5.05} \\
57 \quad & 2 \times 2 \times 2 = 8 \cdot = 8a^3 \\
58 \quad & \text{sign: } -; 6 \times 6 \times 2 = 72 \cdot = \\
-72m^2y
\end{aligned}$$

$$\begin{aligned}
59 \quad & \frac{4}{9} \text{ 4.03} \\
60 \quad & 3 \times 3 = 9 \cdot = 9a^2 \\
61 \quad & 7c + 21 \text{ 5.02} \\
62 \quad & 4 \times 4 \times 4 = 64 \cdot = 64b^3 \\
63 \quad & \text{sign: } +; 3 \times 4 \times 3 = 36 \cdot = \\
36m^2x \\
64 \quad & 4k \text{ 5.04} \\
65 \quad & 6 \times 6 = 36 \cdot = 36m^2 \\
66 \quad & = 6k + 3k + 2m - 7m \cdot = 9k \\
- 5m \text{ 5.07}
\end{aligned}$$

5.09 Dividing terms (booklet p. 17)

Examples (model these live)

$$\begin{aligned}
E1 \quad & 15x \div 3 \\
15 \div 3 = 5 \cdot & = 5x
\end{aligned}$$

$$\begin{aligned}
E2 \quad & 12ab \div 4a \\
= \frac{12ab}{4a} \cdot & = 3b
\end{aligned}$$

$$\begin{aligned}
E3 \quad & \frac{18m^2}{6m} \\
18 \div 6 = 3, m^2 \div m = m \cdot & = 3m
\end{aligned}$$

Questions (49 on today's skill, 17 from earlier lessons)

$$\begin{aligned}
1 \quad & 7p \\
2 \quad & x \\
3 \quad & 5 \\
4 \quad & 4y \\
5 \quad & n \\
6 \quad & 12 \\
7 \quad & 5m \\
8 \quad & c \\
9 \quad & 12bc \text{ 5.08} \\
10 \quad & 2 \\
11 \quad & 6b \\
12 \quad & t \\
13 \quad & 11 \\
14 \quad & 8t \\
15 \quad & y \\
16 \quad & 10 \\
17 \quad & x \\
18 \quad & 3 \\
19 \quad & 4m \\
20 \quad & 15; \text{ yes } \text{ 3.04}
\end{aligned}$$

$$\begin{aligned}
21 \quad & = \frac{32mp}{4p} \cdot = 8m \\
22 \quad & m \\
23 \quad & 7 \\
24 \quad & 4a \text{ 5.07} \\
25 \quad & 3n \\
26 \quad & = \frac{12ab}{3b} \cdot = 4a \\
27 \quad & p \\
28 \quad & = \frac{24mn}{4m} \cdot = 6n \\
29 \quad & 9 \div 3 = 3 \cdot = 3p \\
30 \quad & = 23 + 77 + 14 + 40 \cdot = 100 \\
+ 54 \cdot & = 154 \text{ 5.01} \\
31 \quad & \text{numbers: } 6 \div 3 = 2; \text{ letters: } \\
k \cdot & = 2k \\
32 \quad & = \frac{10by}{2y} \cdot = 5b \\
33 \quad & P = 2 \times 13 + 2 \times 9 \cdot P = 44 \\
\text{5.05} \\
34 \quad & 36 \div 4 = 9 \cdot = 9b \\
35 \quad & \text{numbers: } 10 \div 2 = 5; \\
\text{letters: } m \cdot & = 5m \\
36 \quad & 10 \div 5 = 2 \cdot = 2t
\end{aligned}$$

$$\begin{aligned}
37 \quad & \text{numbers: } 9 \div 3 = 3; \text{ letters: } \\
m \cdot & = 3m \\
38 \quad & \frac{1}{4} \text{ 4.02} \\
39 \quad & = \frac{8bn}{2n} \cdot = 4b \\
40 \quad & 7 \times 14 = 98 \cdot 70 + 28 = 98 \\
\text{5.02} \\
41 \quad & \text{numbers: } 6 \div 2 = 3 \cdot \\
\text{letters: } c^2y \div cy = c \cdot & = 3c \\
42 \quad & 6 \div 3 = 2 \cdot = 2t \\
43 \quad & 3c^2 \text{ 5.08} \\
44 \quad & \text{numbers: } 15 \div 5 = 3; \\
\text{letters: } k \cdot & = 3k \\
45 \quad & = 3c + 5c + 4 \cdot = 8c + 4 \text{ 5.07} \\
46 \quad & = \frac{18ax}{2x} \cdot = 9a \\
47 \quad & \text{numbers: } 20 \div 4 = 5 \cdot \\
\text{letters: } p^2x \div px = p \cdot & = 5p \\
48 \quad & \text{width} = 32py \div 4p \cdot = 8y \\
49 \quad & 5 + 2 = 7 \cdot \frac{5}{10} + \frac{4}{10} = \frac{9}{10} \cdot = \\
7\frac{9}{10} \text{ 4.04}
\end{aligned}$$

$$\begin{aligned}
50 \quad & \text{different signs: } - \cdot = -3 \\
51 \quad & 4 \text{ 5.05} \\
52 \quad & \text{numbers: } 18 \div 6 = 3 \cdot \\
\text{letters: } k^2t \div kt = k \cdot & = 3k \\
53 \quad & \text{width} = 63at \div 7a \cdot = 9t \\
54 \quad & 2 \times k \times p \text{ 5.03} \\
55 \quad & \text{different signs: } - \cdot = -5 \\
56 \quad & = 3 \times 2 \times ck \times k \cdot = 6ck^2 \text{ 5.08} \\
57 \quad & \text{width} = 27xy \div 9x \cdot = 3y \\
58 \quad & \text{different signs: } - \cdot = -4 \\
59 \quad & \frac{2}{3} \text{ 4.12} \\
60 \quad & \text{numbers: } 30 \div 6 = 5 \cdot \\
\text{letters: } c^2n \div cn = c \cdot & = 5c \\
61 \quad & b - 2 \text{ 5.04} \\
62 \quad & \text{numbers: } 8 \div 4 = 2; \text{ letters: } \\
c \cdot & = 2c \\
63 \quad & \text{width} = 48at \div 6a \cdot = 8t \\
64 \quad & 20k \text{ 5.07} \\
65 \quad & \text{different signs: } - \cdot = -5 \\
66 \quad & = 5 \times 4 \cdot = 20 \text{ 5.05}
\end{aligned}$$